

EXERCISE 14

Solution 1: It is given that $b = 24$ cm and $h = 14.5$ cm

$$\begin{aligned}\therefore \text{Area of triangle} &= \frac{1}{2} \times b \times h \\ &= \frac{1}{2} \times 24 \times 14.5 \\ &= 174 \text{ cm}^2\end{aligned}$$

Therefore, the area of the triangle is 174 cm^2 .

Solution 2: Take x be the altitude (height)

$\therefore$ The base of the triangular field is $3x$

Now,

$$\begin{aligned}\text{Area of triangle} &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= \frac{1}{2} \times x \times 3x \\ &= \frac{3}{2} x^2\end{aligned}$$

Cost of sowing the field per hectare = ₹ 58

Total rate of sowing the field = ₹ 783

$\therefore$ Total cost = Area of the field $\times$ ₹ 58

$$\Rightarrow \frac{3}{2} x^2 \times \left(\frac{58}{10000} \right) = 783 \quad [1 \text{ hectare} = 10,000 \text{ sq. metre}]$$

$$\Rightarrow x^2 = \left(\frac{783}{58} \right) \times \left(\frac{2}{3} \right) \times 10000$$

$$\Rightarrow x^2 = 90000$$

$$\Rightarrow x = \sqrt{90000}$$

$$\Rightarrow x = 300 \text{ m}$$

$$\therefore \text{Base} = 3 \times 300 = 900 \text{ m}$$

Therefore, base = 900 m and height = 300 m.

Solution 3: Consider $a = 42$ cm, $b = 34$ cm and $c = 20$ cm

$$s = \frac{a + b + c}{2}$$

$$\Rightarrow s = \frac{42 + 34 + 20}{2} = 48 \text{ cm}$$

Now,

$$\text{Area of the triangle} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{48(48-42)(48-34)(48-20)}$$

$$= \sqrt{48 \times 6 \times 14 \times 28}$$

$$= \sqrt{6 \times 8 \times 6 \times 14 \times 14 \times 2}$$

$$= 6 \times 4 \times 14$$

$$= 336 \text{ cm}^2$$

Given: Longest side $b = 42$ cm

Consider h as the height corresponding to the longest side

$$\text{Area of the triangle} = \frac{1}{2} \times b \times h = \frac{1}{2} \times 42 \times h$$

$$\Rightarrow \frac{1}{2} \times 42 \times h = 336$$

$$\Rightarrow 42 \times h = 336 \times 2$$

$$\Rightarrow h = \frac{336 \times 2}{42}$$

$$\Rightarrow h = 16 \text{ cm}$$

Therefore, Area = 336 cm^2 and the height corresponding to the longest side is 16 cm.

Solution 4:

It is given that $a = 18$ cm, $b = 24$ cm and $c = 30$ cm

$$s = \frac{a + b + c}{2}$$

$$\Rightarrow s = \frac{18 + 24 + 30}{2} = 36 \text{ cm}$$

Now,

$$\text{Area of the triangle} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{36(36 - 18)(36 - 24)(36 - 30)}$$

$$= \sqrt{36 \times 18 \times 12 \times 6}$$

$$= \sqrt{6 \times 6 \times 6 \times 3 \times 3 \times 4 \times 6}$$

$$= 6 \times 6 \times 3 \times 2$$

$$= 216 \text{ cm}^2$$

Given: Smallest side $b = 18 \text{ cm}$

Consider h as the height corresponding to the smallest side

$$\therefore \text{Area of the triangle} = \frac{1}{2} \times b \times h$$

$$\Rightarrow \frac{1}{2} \times b \times h = 216$$

$$\Rightarrow 18 \times h = 216 \times 2$$

$$\Rightarrow h = \frac{216 \times 2}{18}$$

$$\Rightarrow h = 24 \text{ cm}$$

Therefore, Area = 216 cm^2 and the length of the altitude corresponding to the smallest side is 24 cm.

Solution 5:

Consider $a = 91\text{m}$, $b = 98\text{m}$ and $c = 105\text{m}$

$$s = \frac{a + b + c}{2}$$

$$\Rightarrow s = \frac{91 + 98 + 105}{2}$$

$$\Rightarrow s = 147 \text{ m}$$

Now,

$$\text{Area of the triangle} = \sqrt{s(s - a)(s - b)(s - c)}$$

$$= \sqrt{147(147 - 91)(147 - 98)(147 - 105)}$$

$$= \sqrt{147 \times 56 \times 49 \times 42}$$

$$= \sqrt{49 \times 3 \times 7 \times 2 \times 2 \times 2 \times 49 \times 7 \times 3 \times 2}$$

$$= 49 \times 3 \times 2 \times 2 \times 7$$

$$= 4116 \text{ m}^2$$

Given: Longest side $b = 105 \text{ m}$

Consider h as the height corresponding to the longest side

$$\therefore \text{Area of the triangle} = \frac{1}{2} \times b \times h$$

$$\Rightarrow \frac{1}{2} \times b \times h = 4116$$

$$\Rightarrow 105 \times h = 4116 \times 2$$

$$\Rightarrow h = \frac{4116 \times 2}{105}$$

$$\Rightarrow h = 78.4 \text{ m}$$

Therefore, Area = 4116 m^2 and the height corresponding to the longest sides is 78.4 m .

Solution 6: Consider the sides of the triangle as $5x$, $12x$ and $13x$

It is given that perimeter of the triangle = 150m

$$\therefore 5x + 12x + 13x = 150\text{m}$$

$$\Rightarrow 30x = 150$$

$$\Rightarrow x = \frac{150}{30}$$

$$\Rightarrow x = 5 \text{ m}$$

By substituting the value of x we get

$$5x = 5 \times 5 = 25 \text{ m}$$

$$12x = 12 \times 5 = 60 \text{ m}$$

$$13x = 13 \times 5 = 65 \text{ m}$$

Consider $a = 25\text{m}$, $b = 60\text{m}$ and $c = 65\text{m}$

$$s = \frac{a + b + c}{2}$$

$$\Rightarrow s = \frac{25 + 60 + 65}{2} = 75\text{m}$$

Now,

$$\text{Area of triangle} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\begin{aligned} &= \sqrt{75(75 - 25)(75 - 60)(75 - 65)} \\ &= \sqrt{75 \times 50 \times 15 \times 10} \\ &= \sqrt{25 \times 3 \times 25 \times 2 \times 5 \times 3 \times 5 \times 2} \\ &= 25 \times 3 \times 2 \times 5 \\ &= 750 \text{ m}^2 \end{aligned}$$

Therefore, the area of the triangle is 750 sq. m.

Solution 7: Consider a, b, c as the sides of a triangle in the ratio 25: 17: 12

Let, $a = 25, b = 17$ and $c = 12$

$$\therefore \text{Perimeter} = 540 \text{ m}$$

$$\Rightarrow 25x + 17x + 12x = 540$$

$$\Rightarrow 54x = 540$$

$$\Rightarrow x = 10$$

By substituting the value of x

$$a = 25x = 25 \times 10 = 250\text{m}$$

$$b = 17x = 17 \times 10 = 170\text{m}$$

$$c = 12x = 12 \times 10 = 120\text{m}$$

We know that

$$s = \frac{a + b + c}{2}$$

$$\Rightarrow s = \frac{250 + 170 + 120}{2} = 270\text{m}$$

Now,

$$\text{Area of the triangle} = \sqrt{s(s - a)(s - b)(s - c)}$$

$$= \sqrt{270(270 - 250)(270 - 170)(270 - 120)}$$

$$= \sqrt{270 \times 20 \times 100 \times 150}$$

$$= \sqrt{9 \times 6 \times 5 \times 4 \times 5 \times 100 \times 25 \times 6}$$

$$= 3 \times 6 \times 5 \times 2 \times 10 \times 5$$

$$= 9000 \text{ m}^2$$

The cost of ploughing the field is ₹ 5 per m^2

$\therefore$ The cost of ploughing $9000 \text{ m}^2 = ₹5 \times 9000 = ₹45000$

Therefore, the area of the field is 9000 m^2 and the cost of ploughing the field is ₹ 45000.

Solution 8:

(i) It is given that two sides of a triangular field are 85m and 154m

Consider the third side as $x \text{ m}$

It is given that the perimeter = 324m

$$85 + 154 + x = 324$$

$$\Rightarrow x = 324 - 85 - 154$$

$$\Rightarrow x = 85 \text{ m}$$

Consider $a = 85 \text{ m}$, $b = 154 \text{ m}$ and $c = 85 \text{ m}$

$$s = \frac{a + b + c}{2}$$

$$\Rightarrow s = \frac{85 + 154 + 85}{2}$$

$$\Rightarrow s = \frac{324}{2}$$

$$\Rightarrow s = 162 \text{ m}$$

Now,

$$\begin{aligned} \text{Area of the triangle} &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{162(162-85)(162-154)(162-85)} \\ &= \sqrt{162(77)(8)(77)} \\ &= \sqrt{1296 \times 77 \times 77} \\ &= \sqrt{36 \times 36 \times 77 \times 77} \\ &= 36 \times 77 \\ &= 2772 \text{ m}^2 \end{aligned}$$

(ii) We can find out the height of the triangle corresponding to 154 m in the following manner:

We have:

$$\text{Area of triangle} = 2772 \text{ m}^2$$

$$\Rightarrow \frac{1}{2} \times \text{Base} \times \text{Height} = 2772$$

$$\Rightarrow \frac{1}{2} \times 154 \times \text{Height} = 2772$$

$$\Rightarrow \text{Height} = \frac{2772 \times 2}{154}$$

$$\Rightarrow \text{Height} = 36 \text{ m}$$

Solution 9: Let $a = 13 \text{ cm}$, $b = 13 \text{ cm}$ and $c = 20 \text{ cm}$

Now,

$$s = \frac{1}{2}(a + b + c) = \frac{13 + 13 + 20}{2} = \frac{46}{2} = 23 \text{ cm}$$

$$\begin{aligned} \therefore \text{area of the triangle} &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{23(23-13)(23-13)(23-20)} \\ &= \sqrt{23 \times 10 \times 10 \times 3} \\ &= 10\sqrt{69} \\ &= 10 \times 8.306 \\ &= 83.06 \text{ cm}^2 \end{aligned}$$

Therefore, area of an isosceles triangle is 83.06 cm^2 .

Solution 10: Consider $\triangle ABC$ as an isosceles triangle with AL perpendicular to BC

It is given that $BC = 80 \text{ cm}$ and $\text{area} = 360 \text{ cm}^2$

We know that,

$$\text{area of a triangle} = \frac{1}{2} \times b \times h$$

Now,

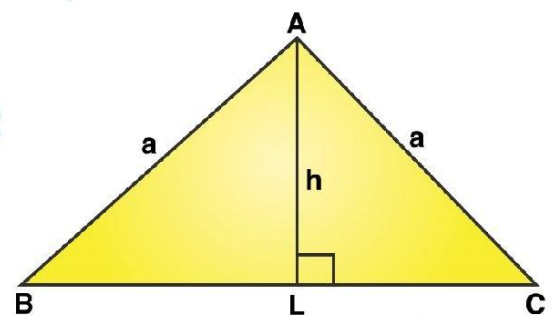
$$\frac{1}{2} \times BC \times AL = 360 \text{ cm}^2$$

$$\Rightarrow \frac{1}{2} \times 80 \times AL = 360 \text{ cm}^2$$

$$\Rightarrow 40 \times AL = 360$$

$$\Rightarrow AL = 9 \text{ cm}$$

Now, $BL = \frac{1}{2} \times BC$



$$\Rightarrow BL = \frac{1}{2} \times 80$$

$$\Rightarrow BL = 40 \text{ cm}$$

Using the Pythagoras theorem

We know that

$$a = \sqrt{(BL)^2 + AL^2}$$

$$\Rightarrow a = \sqrt{(40)^2 + 9^2}$$

$$\Rightarrow a = \sqrt{(1600 + 81)}$$

$$\Rightarrow a = \sqrt{1681}$$

$$\Rightarrow a = 41 \text{ cm}$$

So, the perimeter of the isosceles triangle = $41 + 41 + 80 = 162 \text{ cm}$

Therefore, the perimeter of the triangle is 162 cm .

Solution 11: The ratio of the equal side to its base is $3 : 2$.

$$\Rightarrow \text{Ratio of sides} = 3 : 3 : 2.$$

Let the three sides of triangle be $3x, 3x, 2x$.

The perimeter of isosceles triangle = 32 cm .

$$\Rightarrow 3x + 3x + 2x = 32 \text{ cm}$$

$$\Rightarrow 8x = 32$$

$$\Rightarrow x = 4 \text{ cm}$$

Therefore, the three side of triangle are $3x, 3x, 2x = 12 \text{ cm}, 12 \text{ cm}, 8 \text{ cm}$.

Let S be the semi-perimeter of the triangle.

Then,

$$S = \frac{1}{2}(12 + 12 + 8) = \frac{32}{2} = 16$$

$$\begin{aligned}\therefore \text{Area of the triangle} &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{16(16-12)(16-12)(16-8)} \\ &= \sqrt{16 \times 4 \times 4 \times 8}\end{aligned}$$

$$= \sqrt{16 \times 16 \times 8}$$

$$= \sqrt{16 \times 16 \times 4 \times 2}$$

$$= 4 \times 4 \times 2\sqrt{2}$$

$$= 32\sqrt{2} \text{ cm}^2$$

Solution 12: Let ABC be any triangle with perimeter 50 cm.

Let the smallest side of the triangle be x.

Then the other sides be $x + 4$ and $2x - 6$.

Now,

$$x + x + 4 + 2x - 6 = 50 \quad (\because \text{perimeter is } 50 \text{ cm})$$

$$\Rightarrow 4x - 2 = 50$$

$$\Rightarrow 4x = 50 + 2$$

$$\Rightarrow 4x = 52$$

$$\Rightarrow x = 13$$

So, the sides of the triangle are

$$a = x + 4 = 13 + 4 = 17 \text{ cm}$$

$$b = 2x - 6 = 2(13) - 6 = 20 \text{ cm}$$

$$c = x = 13 \text{ cm}$$

$\therefore$ Semi-perimeter of the triangle is

$$s = \frac{a + b + c}{2} = \frac{13 + 17 + 20}{2} = \frac{50}{2} = 25 \text{ cm}$$

$$\therefore \text{Area of the triangle} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{25(25-17)(25-20)(25-13)}$$

$$= \sqrt{25 \times 8 \times 5 \times 12}$$

$$= \sqrt{25 \times 4 \times 2 \times 5 \times 4 \times 3}$$

$$= 5 \times 2 \times 2\sqrt{2 \times 5 \times 3}$$

$$= 20\sqrt{30} \text{ cm}^2$$

Hence, the area of the triangle is $= 20\sqrt{30} \text{ cm}^2$.

Solution 13: Three sides of a wall are 13 m, 14 m and 15 m respectively.

i.e.,

$$a = 13 \text{ m}, b = 14 \text{ m and } c = 15 \text{ m}$$

∴ Semi-perimeter of the triangle is

$$s = \frac{a + b + c}{2} = \frac{13 + 14 + 15}{2} = \frac{42}{2} = 21 \text{ m}$$

$$\begin{aligned}\therefore \text{Area of the triangle} &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{21(21-13)(21-14)(21-15)} \\ &= \sqrt{21 \times 8 \times 7 \times 6} \\ &= \sqrt{7 \times 3 \times 4 \times 2 \times 7 \times 3 \times 2} \\ &= 7 \times 2 \times 3 \times 2 \\ &= 84 \text{ m}^2\end{aligned}$$

Now,

The rent of advertisements per m^2 per year = ₹ 2000

$$\begin{aligned}\text{The rent of the wall with area } 84 \text{ m}^2 \text{ per year} &= ₹ 2000 \times 84 \\ &= ₹ 168000\end{aligned}$$

$$\begin{aligned}\text{The rent of the wall with area } 84 \text{ m}^2 \text{ for 6 months} &= ₹ \frac{168000}{2} \\ &= ₹ 84000\end{aligned}$$

Solution 14:

Let the equal sides of an isosceles triangle be x cm each

$$\therefore \text{Base of the triangle} = \frac{3}{2}x \text{ cm}$$

(i) perimeter = 42cm

$$x + x + \frac{3}{2}x = 42$$

$$\Rightarrow 2x + \frac{3}{2}x = 42$$

$$\Rightarrow \frac{4x + 3x}{2} = 42$$

$$\Rightarrow \frac{7x}{2} = 42$$

$$\Rightarrow x = 42 \times \frac{2}{7}$$

$$\Rightarrow x = 12 \text{ cm}$$

So, equal sides of the triangle are 12 cm each.

$$\therefore \text{Base} = \frac{3}{2}x = \frac{3}{2} \times 12 = 18 \text{ cm}$$

Therefore, the length of each side of the triangle is 12cm, 12 cm and 18cm.

(ii) Consider $a = 12 \text{ cm}$, $b = 12 \text{ cm}$ and $c = 18 \text{ cm}$

$$s = \frac{a + b + c}{2} = \frac{12 + 12 + 18}{2} = 21$$

$$\therefore \text{area of triangle} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{21(21-12)(21-12)(21-18)}$$

$$= \sqrt{21 \times 9 \times 9 \times 3}$$

$$= \sqrt{3 \times 7 \times 9 \times 9 \times 3}$$

$$= 9 \times 3\sqrt{7}$$

$$= 27 \times 2.64$$

$$= 71.28 \text{ cm}^2$$

Therefore, the area of the triangle is 71.28 cm^2 .

(iii) Area of a triangle = 71.28 cm^2

$$\Rightarrow \frac{1}{2} \times \text{base} \times \text{height} = 71.28$$

$$\Rightarrow \frac{1}{2} \times 18 \times \text{height} = 71.28$$

$$\Rightarrow 9 \times \text{height} = 71.28$$

$$\Rightarrow \text{height} = \frac{71.28}{9}$$

$$\Rightarrow \text{height} = 7.92 \text{ cm}$$

Solution 15: Let, each side of the triangle be a cm

$$\text{Area of equilateral triangle} = 36\sqrt{3} \text{ cm}^2$$

$$\therefore \frac{\sqrt{3}}{4} \times a^2 = 36\sqrt{3}$$

$$\Rightarrow a^2 = \frac{36\sqrt{3} \times 4}{\sqrt{3}}$$

$$\Rightarrow a^2 = 36 \times 4$$

$$\Rightarrow a^2 = 144$$

$$\Rightarrow a = \sqrt{144}$$

$$\therefore a = 12 \text{ cm}$$

$$\therefore \text{Perimeter} = 3a = 3 \times 12 = 36 \text{ cm}$$

Solution 16:

Let, each side of the triangle be a cm

$$\text{Area of equilateral triangle} = 81\sqrt{3} \text{ cm}^2$$

$$\therefore \frac{\sqrt{3}}{4} \times a^2 = 81\sqrt{3}$$

$$\Rightarrow a^2 = \frac{81\sqrt{3} \times 4}{\sqrt{3}}$$

$$\Rightarrow a^2 = 81 \times 4$$

$$\Rightarrow a^2 = (9 \times 2)^2$$

$$\Rightarrow a = 9 \times 2$$

$$\therefore a = 18 \text{ cm}$$

$$\therefore \text{Height} = \frac{\sqrt{3}}{2} \times \text{side}$$

$$= \left(\frac{\sqrt{3}}{2} \times 18 \right) \text{ cm}$$

$$= 9\sqrt{3} \text{ cm}$$

Solution 17: Each side of an equilateral triangle be 8 cm

$$\text{Area of equilateral triangle} = \frac{\sqrt{3}}{4} \times (\text{side})^2$$

$$= \frac{\sqrt{3}}{4} \times (8)^2$$

$$= \frac{\sqrt{3}}{4} \times 64$$

$$= 16\sqrt{3}$$

$$= 16 \times 1.732$$

$$= 27.71 \text{ cm}^2$$

$$\text{(ii) Height of an equilateral triangle} = \frac{\sqrt{3}}{2} \times \text{side}$$

$$= \left(\frac{\sqrt{3}}{2} \times 8 \right) \text{ cm}$$

$$= 4\sqrt{3} \text{ cm}$$

$$= 4 \times 1.732 \text{ cm}$$

$$= 6.928 \text{ cm}$$

$$= 6.93 \text{ cm}$$

Solution 18:

Height of an equilateral triangle = 9 cm

$$\therefore \frac{\sqrt{3}}{2} \times \text{side} = 9$$

$$\Rightarrow \text{side} = \frac{9 \times 2}{\sqrt{3}}$$

$$\Rightarrow \text{side} = \frac{3 \times \sqrt{3} \times \sqrt{3} \times 2}{\sqrt{3}}$$

$$\Rightarrow \text{side} = 6\sqrt{3} \text{ cm}$$

$$\text{Also, Area of equilateral triangle} = \frac{\sqrt{3}}{4} \times (\text{side})^2$$

$$= \frac{\sqrt{3}}{4} \times (6\sqrt{3})^2$$

$$= \frac{\sqrt{3}}{4} \times 36 \times 3$$

$$= 27\sqrt{3}$$

$$= 27 \times 1.732$$

$$= 46.76 \text{ cm}^2$$

Solution 19: Base = BC = 48 cm

Hypotenuse = AC = 50 cm

Let, height = AB = x cm

Using the Pythagoras theorem,

$$AC^2 = AB^2 + BC^2$$

$$\Rightarrow 50^2 = x^2 + 48^2$$

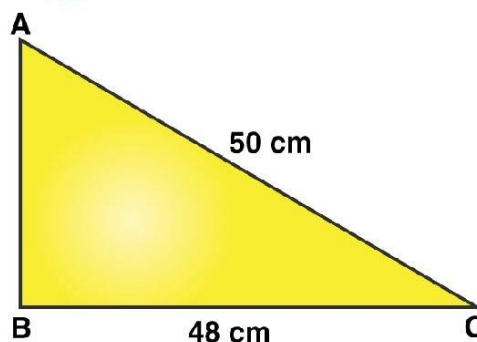
$$\Rightarrow x^2 = 50^2 - 48^2$$

$$\Rightarrow x^2 = 2500 - 2304$$

$$\Rightarrow x^2 = 196$$

$$\Rightarrow x = \sqrt{196}$$

$$\Rightarrow x = 14 \text{ cm}$$



Area of a right angled triangle = $\left(\frac{1}{2} \times b \times h\right)$

$$= \left(\frac{1}{2} \times 48 \times 14\right) \text{ cm}^2$$

$$= 336 \text{ cm}^2$$

Solution 20: In right angled $\triangle ABD$,

$$AB^2 = AD^2 + DB^2 \quad (\text{Pythagoras Theorem})$$

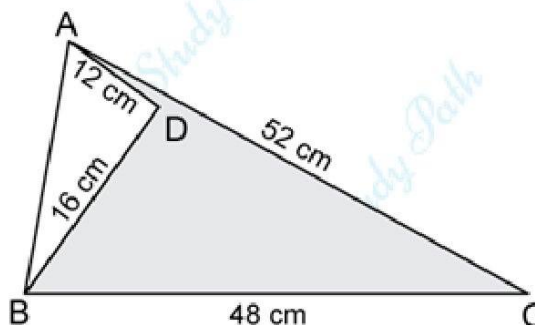
$$\Rightarrow AB^2 = 12^2 + 16^2$$

$$\Rightarrow AB^2 = 144 + 256$$

$$\Rightarrow AB^2 = 400$$

$$\Rightarrow AB = \sqrt{400}$$

$$\Rightarrow AB = 20 \text{ cm}$$



$$\begin{aligned}\therefore \text{Area of ABD} &= \frac{1}{2} \times b \times h \\ &= \frac{1}{2} \times BD \times AD \\ &= \frac{1}{2} \times 16 \times 12 \\ &= 96 \text{ cm}^2\end{aligned}$$

In $\triangle ABC$, sides are 20 cm, 52 cm and 48 cm

$$\begin{aligned}\therefore \text{semi perimeter, } s &= \frac{a + b + c}{2} \\ &= \frac{20 + 52 + 48}{2} \\ &= \frac{120}{2} \\ &= 60 \text{ cm}\end{aligned}$$

$$\begin{aligned}\therefore \text{Area of } \triangle ACB &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{60(60-20)(60-52)(60-48)} \\ &= \sqrt{60 \times 40 \times 8 \times 12} \\ &= \sqrt{12 \times 5 \times 5 \times 8 \times 8 \times 12} \\ &= 12 \times 5 \times 8 \\ &= 480 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Now, area of the shaded region} &= \text{Area of } \triangle ABC - \text{Area of } \triangle ABD \\ &= 480 - 96 \\ &= 384 \text{ cm}^2\end{aligned}$$

Therefore, the area of the shaded region is 384 cm^2 .

Solution 21: Consider ABCD as a quadrilateral

It is given that $AB = 6\text{cm}$, $BC = 8\text{cm}$, $CD = 12\text{cm}$ and $AD = 14\text{cm}$

Consider $\triangle ABC$

Using the Pythagoras theorem

$$AC^2 = AB^2 + BC^2$$

$$\Rightarrow AC^2 = 6^2 + 8^2$$

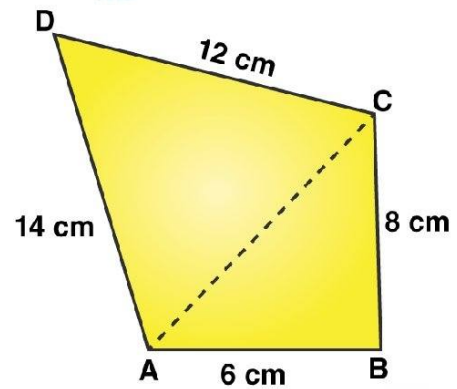
$$\Rightarrow AC^2 = 36 + 64$$

$$\Rightarrow AC^2 = 100$$

$$\Rightarrow AC = \sqrt{100}$$

$$\Rightarrow AC = 10 \text{ cm}$$

$$\begin{aligned}\therefore \text{Area of } \triangle ABC &= \frac{1}{2} \times b \times h \\ &= \frac{1}{2} \times AB \times BC \\ &= \frac{1}{2} \times 6 \times 8 \\ &= 24 \text{ cm}^2\end{aligned}$$



Let $a = 10 \text{ cm}$, $b = 12 \text{ cm}$ and $c = 14 \text{ cm}$

$$\begin{aligned}\therefore \text{semi perimeter, } s &= \frac{a + b + c}{2} \\ &= \frac{10 + 12 + 14}{2} \\ &= \frac{36}{2} \\ &= 18 \text{ cm}\end{aligned}$$

$$\begin{aligned}\therefore \text{Area of } \triangle ACD &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{18(18-10)(18-12)(18-14)} \\ &= \sqrt{18 \times 8 \times 6 \times 4} \\ &= \sqrt{9 \times 2 \times 2 \times 4 \times 6 \times 4} \\ &= 3 \times 2 \times 4\sqrt{6} \\ &= 24\sqrt{6} \\ &= 24 \times 2.45 \\ &= 58.8 \text{ cm}^2\end{aligned}$$

Thus,

$$\begin{aligned}\text{Area of quadrilateral ABCD} &= \text{Area of } \triangle ABC + \text{Area of } \triangle ACD \\ &= (24 + 58.8) \text{ cm}^2 \\ &= 82.8 \text{ cm}^2\end{aligned}$$

Hence, the area of quadrilateral ABCD is 82.8 cm².

Solution 22:

In $\triangle ABD$

$$AD^2 = AB^2 + BD^2 \quad [\text{Using the Pythagoras theorem}]$$

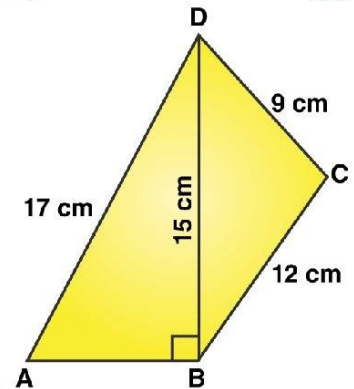
$$\Rightarrow (17)^2 = AB^2 + (15)^2$$

$$\Rightarrow 289 = AB^2 + 225$$

$$\Rightarrow AB^2 = 289 - 225$$

$$\Rightarrow AB^2 = 64$$

$$\Rightarrow AB = 8 \text{ cm}$$



$$\begin{aligned}\therefore \text{Perimeter of quadrilateral ABCD} &= AB + BC + CD + AD \\ &= 8 + 12 + 9 + 17 \\ &= 46 \text{ cm}\end{aligned}$$

Now,

$$\begin{aligned}\text{Area of } \triangle ABD &= \frac{1}{2} \times b \times h \\ &= \frac{1}{2} \times AB \times BD \\ &= \frac{1}{2} \times 8 \times 15 \\ &= 60 \text{ cm}^2\end{aligned}$$

Consider $\triangle BCD$

We know that $BC = 12 \text{ cm}$, $CD = 9 \text{ cm}$ and $BD = 15 \text{ cm}$

It can be written as $a = 12 \text{ cm}$, $b = 9 \text{ cm}$ and $c = 15 \text{ cm}$

$$\begin{aligned}\therefore \text{semi perimeter, } s &= \frac{a + b + c}{2} \\ &= \frac{12 + 9 + 15}{2}\end{aligned}$$

$$= \frac{36}{2}$$

$$= 18 \text{ cm}$$

$$\therefore \text{Area of } \triangle BCD = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{18(18-12)(18-9)(18-15)}$$

$$= \sqrt{18 \times 6 \times 9 \times 3}$$

$$= \sqrt{6 \times 3 \times 6 \times 9 \times 3}$$

$$= 6 \times 3 \times 3$$

$$= 54 \text{ cm}^2$$

So, the area of quadrilateral ABCD = Area of $\triangle ABD$ + Area of $\triangle BCD$

$$= 60 + 54$$

$$= 114 \text{ cm}^2$$

Therefore, the perimeter is 46 cm and the area is 114 cm².

Solution 23: In $\triangle BAC$

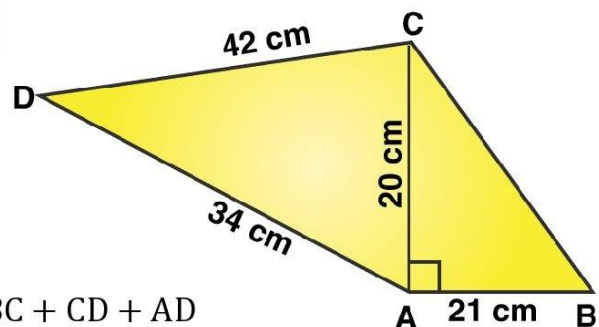
$$BC^2 = AC^2 + AB^2 \quad [\text{Using the Pythagoras theorem}]$$

$$\Rightarrow BC^2 = 20^2 + 21^2$$

$$\Rightarrow BC^2 = 400 + 441$$

$$\Rightarrow BC^2 = 841$$

$$\Rightarrow BC = 29 \text{ cm}$$



$$\therefore \text{Perimeter of quadrilateral ABCD} = AB + BC + CD + AD$$

$$= 21 + 29 + 42 + 34$$

$$= 126 \text{ cm}$$

Now,

$$\text{Area of } \triangle ABC = \frac{1}{2} \times b \times h$$

$$= \frac{1}{2} \times AB \times AC$$

$$= \frac{1}{2} \times 21 \times 20$$

$$= 210 \text{ cm}^2$$

In $\triangle ACD$, $AC = 20$ cm, $CD = 42$ cm and $AD = 34$ cm

Let $a = 20$ cm, $b = 42$ cm and $c = 34$ cm

$$\begin{aligned}\therefore \text{semi perimeter, } s &= \frac{a + b + c}{2} \\ &= \frac{20 + 42 + 34}{2} \\ &= \frac{96}{2} \\ &= 48 \text{ cm}\end{aligned}$$

$$\begin{aligned}\therefore \text{Area of } \triangle ACD &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{48(48-20)(48-42)(48-34)} \\ &= \sqrt{48 \times 28 \times 6 \times 14} \\ &= \sqrt{16 \times 3 \times 14 \times 2 \times 3 \times 2 \times 14} \\ &= 4 \times 3 \times 14 \times 2 \\ &= 336 \text{ cm}^2\end{aligned}$$

So the area of quadrilateral ABCD = Area of $\triangle ABC$ + Area of $\triangle ACD$

$$= 210 + 336$$

$$= 546 \text{ cm}^2$$

Solution 24: In $\triangle ABD$

Using the Pythagoras theorem

$$BD^2 = AB^2 + AD^2$$

$$\Rightarrow (26)^2 = AB^2 + (24)^2$$

$$\Rightarrow 676 = AB^2 + 576$$

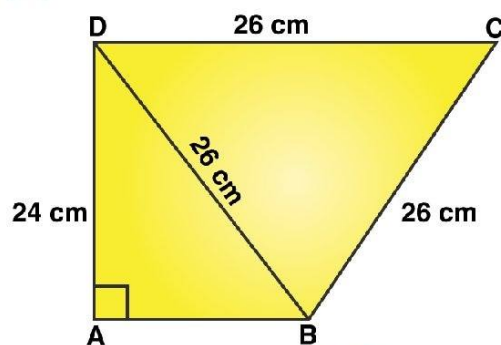
$$\Rightarrow AB^2 = 676 - 576$$

$$\Rightarrow AB^2 = 100$$

$$\Rightarrow AB = 10 \text{ cm}$$

$$\Rightarrow \text{Base} = 10 \text{ cm}$$

Now,



$$\begin{aligned}\text{Area of } \triangle ABD &= \frac{1}{2} \times b \times h \\ &= \frac{1}{2} \times 10 \times 24 \\ &= 120 \text{ cm}^2\end{aligned}$$

Also, we know that $\triangle BDC$ is an equilateral triangle.

$$\begin{aligned}\therefore \text{Area of equilateral triangle} &= \frac{\sqrt{3}}{4} \times (\text{side})^2 \\ &= \frac{\sqrt{3}}{4} \times (26)^2 \\ &= \frac{\sqrt{3}}{4} \times 676 \\ &= 169\sqrt{3} \\ &= 169 \times 1.73 \\ &= 292.37\end{aligned}$$

Now,

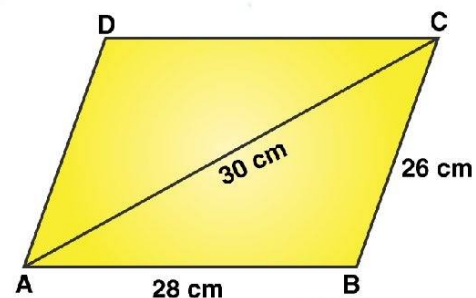
$$\begin{aligned}\text{Area of quadrilateral ABCD} &= \text{Area of } \triangle ABD + \text{Area of } \triangle BDC \\ &= (120 + 292.37) \text{ cm}^2 \\ &= 412.37 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Perimeter of ABCD} &= AB + BC + CD + DA \\ &= 10 + 26 + 26 + 24 \\ &= 86 \text{ cm}\end{aligned}$$

Solution 25: In $\triangle ABC$

Let $a = 26 \text{ cm}$, $b = 30 \text{ cm}$ and $c = 28 \text{ cm}$

$$\begin{aligned}\therefore \text{semi perimeter, } s &= \frac{a + b + c}{2} \\ &= \frac{26 + 30 + 28}{2} \\ &= \frac{84}{2} \\ &= 42 \text{ cm}\end{aligned}$$



$$\begin{aligned}
 \therefore \text{Area of } \triangle ABC &= \sqrt{s(s-a)(s-b)(s-c)} \\
 &= \sqrt{42(42-26)(42-30)(42-28)} \\
 &= \sqrt{42 \times 16 \times 12 \times 14} \\
 &= \sqrt{(14 \times 3) \times (4 \times 4) \times (2 \times 2 \times 3) \times 14} \\
 &= 14 \times 4 \times 2 \times 3 \\
 &= 336 \text{ cm}^2
 \end{aligned}$$

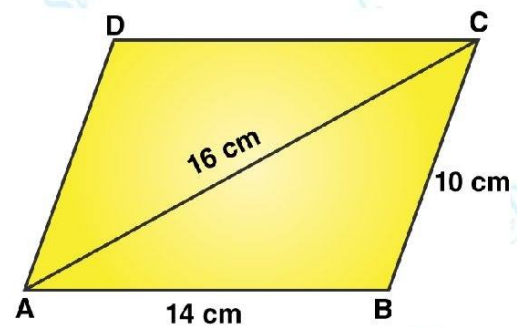
We know that a diagonal divides a parallelogram into two triangles of equal areas.

$$\begin{aligned}
 \therefore \text{Area of parallelogram ABCD} &= 2(\text{Area of triangle ABC}) \\
 &= 2 \times 336 \\
 &= 672 \text{ cm}^2
 \end{aligned}$$

Solution 26: In $\triangle ABC$

Let $a = 10 \text{ cm}$, $b = 16 \text{ cm}$ and $c = 14 \text{ cm}$

$$\begin{aligned}
 \therefore \text{semi perimeter, } s &= \frac{a+b+c}{2} \\
 &= \frac{10+16+14}{2} \\
 &= \frac{40}{2} \\
 &= 20 \text{ cm}
 \end{aligned}$$



$$\begin{aligned}
 \therefore \text{Area of } \triangle ABC &= \sqrt{s(s-a)(s-b)(s-c)} \\
 &= \sqrt{20(20-10)(20-16)(20-14)} \\
 &= \sqrt{20 \times 10 \times 4 \times 6} \\
 &= \sqrt{(10 \times 2) \times 10 \times (2 \times 2) \times (3 \times 2)} \\
 &= 10 \times 2 \times 2\sqrt{3} \\
 &= 69.2 \text{ cm}^2
 \end{aligned}$$

We know that a diagonal divides a parallelogram into two triangles of equal areas.

$$\begin{aligned}\therefore \text{Area of parallelogram ABCD} &= 2(\text{Area of triangle ABC}) \\ &= 2 \times 69.2 \text{ cm}^2 \\ &= 138.4 \text{ cm}^2\end{aligned}$$

Solution 27:

$$\begin{aligned}\text{Area of } \triangle ABD &= \frac{1}{2} \times b \times h \\ &= \frac{1}{2} \times BD \times AL \\ &= \frac{1}{2} \times 64 \times 16.8 \\ &= 537.6 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Area of } \triangle BCD &= \frac{1}{2} \times b \times h \\ &= \frac{1}{2} \times BD \times CM \\ &= \frac{1}{2} \times 64 \times 13.2 \\ &= 422.4 \text{ cm}^2\end{aligned}$$

So,

$$\begin{aligned}\text{Area of quadrilateral ABCD} &= \text{Area of } \triangle ABD + \text{Area of } \triangle BCD \\ &= 537.6 + 422.4 \\ &= 960 \text{ cm}^2\end{aligned}$$

Therefore, the area of quadrilateral ABCD is 960 cm².

Solution 28: Consider x cm as the smaller side of the trapezium

So the larger parallel can be written as (x + 4) cm

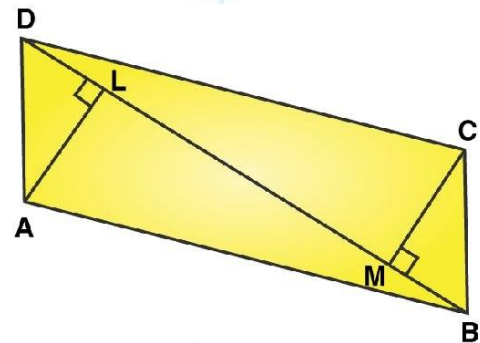
Area of trapezium = $\frac{1}{2} \times \text{sum of parallel sides} \times \text{height}$

$$475 = \frac{1}{2} \times (x + (x + 4)) \times 19$$

$$\Rightarrow 25 = \frac{1}{2} \times (2x + 4)$$

$$\Rightarrow 50 = 2x + 4$$

$$\Rightarrow 2x = 50 - 4$$



$$\Rightarrow 2x = 46$$

$$\Rightarrow x = 23\text{cm}$$

So, we get Larger parallel side $(x + 4) = 23 + 4 = 27\text{cm}$

Therefore, the length of two parallel sides is 23cm and 27cm.

Solution 29: In $\triangle ABC$

It is given that $AB = 7.5\text{ cm}$, $BC = 7\text{ cm}$ and $AC = 6.5\text{ cm}$

Take $a = 7.5\text{ cm}$, $b = 7\text{ cm}$ and $c = 6.5\text{ cm}$

$$\begin{aligned}\therefore \text{semi perimeter, } s &= \frac{a + b + c}{2} \\ &= \frac{7.5 + 6.5 + 7}{2} \\ &= \frac{21}{2} \\ &= 10.5\text{ cm}\end{aligned}$$

$$\begin{aligned}\therefore \text{Area of } \triangle ABC &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{10.5(10.5 - 7.5)(10.5 - 6.5)(10.5 - 7)} \\ &= \sqrt{10.5 \times 3 \times 4 \times 3.5} \\ &= \sqrt{441} \\ &= 21\text{ cm}^2\end{aligned}$$

Now,

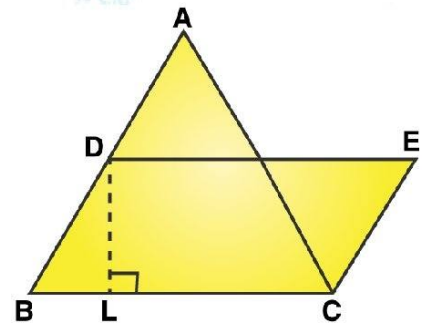
area of parallelogram DBCE = Area of $\triangle ABC$

$$BC \times DL = 21$$

$$\Rightarrow 7 \times DL = 21$$

$$\Rightarrow DL = 3\text{ cm}$$

Therefore, the height DL of the parallelogram is 3cm.



Solution 30:

Construction: Draw $BT \perp CD$

In $\triangle BTC$, by Pythagoras theorem,

$$BT^2 = BC^2 - CT^2$$

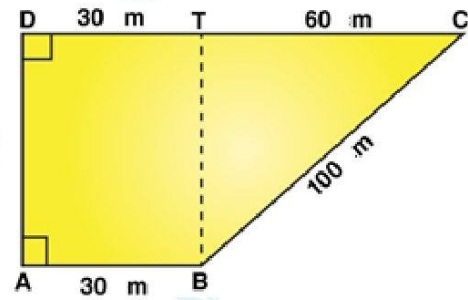
$$\Rightarrow BT^2 = 100^2 - 60^2$$

$$\Rightarrow BT^2 = 10000 - 3600$$

$$\Rightarrow BT^2 = 6400$$

$$\Rightarrow BT = 80 \text{ m}$$

$$\Rightarrow AD = BT = 80 \text{ m}$$



Thus, the height of the trapezium = 80 m

Now, Area of field = Area of trapezium ABCD

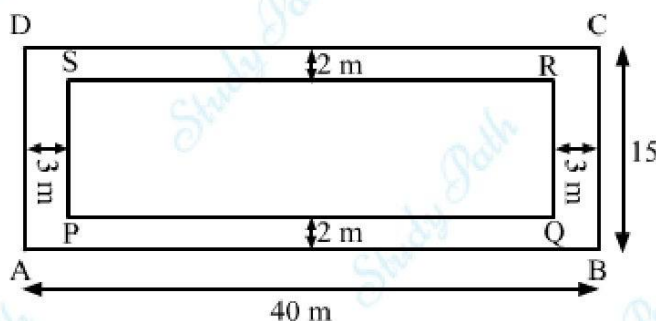
$$\begin{aligned} \therefore \text{Area of trapezium ABCD} &= \frac{1}{2} \times \text{Sum of parallel sides} \times \text{height} \\ &= \frac{1}{2} \times (AB + CD) \times AD \\ &= \frac{1}{2} \times (30 + 90) \times 80 \\ &= 120 \times 40 \\ &= 4800 \text{ m}^2 \end{aligned}$$

It is given that the cost of ploughing 1m^2 is ₹ 5

So, the cost of ploughing $4800 \text{ m}^2 = ₹ (5 \times 4800) = ₹ 24000$

Therefore, the total cost of ploughing the field is ₹ 24000.

Solution 31: Let ABCD be a rectangular plot is given for constructing a house, having a measurement of 40 m long and 15 m in the front.



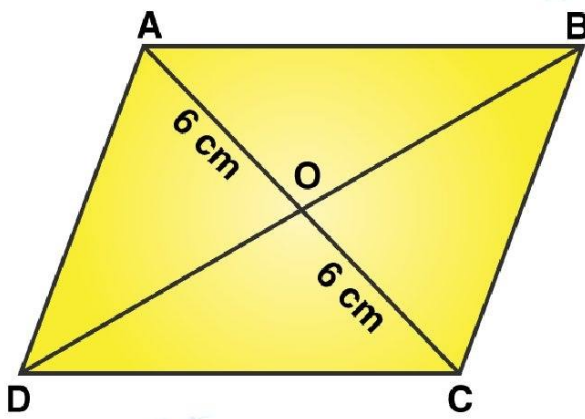
According to the laws, the length of the inner rectangle = $40 - 3 - 3 = 34 \text{ m}$ and

the breath of the inner rectangle = $15 - 2 - 2 = 11$ m.

$$\begin{aligned}\therefore \text{Area of the inner rectangle PQRS} &= \text{Length} \times \text{Breadth} \\ &= 34 \times 11 \\ &= 374 \text{ m}^2\end{aligned}$$

Hence, the largest area where house can be constructed is 374 m^2 .

Solution 32:



Let ABCD as the rhombus shaped sheet

Given,

$$\text{Perimeter} = 40\text{cm}$$

$$4 \times \text{side} = 40$$

$$\text{Side} = 10 \text{ cm}$$

$$\therefore AB = BC = CD = AD = 10 \text{ cm}$$

Let the diagonal AC = 12cm

Since diagonals of a rhombus bisect each other at right angles,

$$AO = OC = 6 \text{ cm}$$

In right $\triangle AOD$, by Pythagoras theorem,

$$OD^2 = AD^2 - AO^2$$

$$\Rightarrow OD^2 = 10^2 - 6^2$$

$$\Rightarrow OD^2 = 100 - 36$$

$$\Rightarrow OD^2 = 64$$

$$\Rightarrow OD = 8 \text{ cm}$$

$$\therefore BD = 2 \times OD = 2 \times 8 = 16 \text{ cm}$$

Now,

We know that

$$\begin{aligned}\text{Area of rhombus ABCD} &= \frac{1}{2} (\text{Product of diagonals}) \\ &= \frac{1}{2} \times AC \times BD \\ &= \frac{1}{2} \times 12 \times 16 \\ &= 96 \text{ cm}^2\end{aligned}$$

It is given that the cost of painting is ₹5 per cm^2

$$\begin{aligned}\text{So, the cost of painting both sides of rhombus} &= ₹ 5 \times (96 + 96) \\ &= ₹ 5 \times 192 \\ &= ₹ 960\end{aligned}$$

Therefore, the cost of painting is ₹960.

Solution 33: Consider a, b and c as the sides of a triangle and s as the semi perimeter

So, we get

$$s - a = 8 \text{ cm}$$

$$s - b = 7 \text{ cm}$$

$$s - c = 5 \text{ cm}$$

Now,

$$(s - a) + (s - b) + (s - c) = 8 + 7 + 5$$

$$\Rightarrow 3s - (a + b + c) = 20$$

$$\Rightarrow 3s - 2s = 20 \quad [\because a + b + c = 2s]$$

$$\Rightarrow s = 20$$

By substituting the value of s

$$a = s - 8 = 20 - 8 = 12 \text{ cm}$$

$$b = s - 7 = 20 - 7 = 13\text{cm}$$

$$c = s - 5 = 20 - 5 = 15\text{cm}$$

$$\begin{aligned}\therefore \text{Area of triangle} &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{20(20-12)(20-13)(20-15)} \\ &= \sqrt{20 \times 8 \times 7 \times 5} \\ &= \sqrt{(4 \times 5) \times (4 \times 2) \times 7 \times 5} \\ &= 4 \times 5\sqrt{14} \\ &= 20\sqrt{14} \text{ cm}^2\end{aligned}$$

Solution 34: Let, $a = 16 \text{ cm}$, $b = 12 \text{ cm}$ and $c = 20 \text{ cm}$

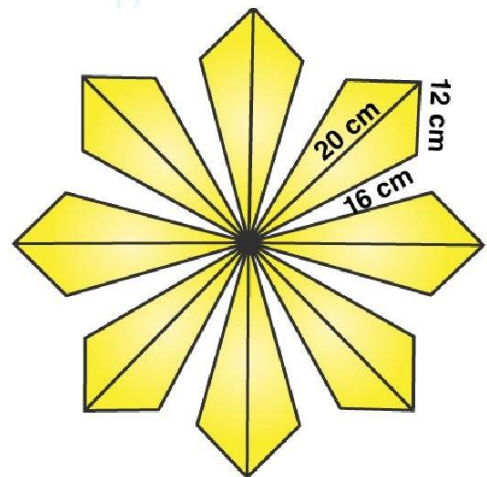
$$\begin{aligned}\therefore \text{semi perimeter, } s &= \frac{a+b+c}{2} \\ &= \frac{16+12+20}{2} \\ &= \frac{48}{2} \\ &= 24 \text{ cm}\end{aligned}$$

$$\begin{aligned}\therefore \text{Area of triangle} &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{24(24-16)(24-12)(24-20)} \\ &= \sqrt{24 \times 8 \times 12 \times 4} \\ &= \sqrt{(8 \times 3) \times 8 \times (3 \times 4) \times 4} \\ &= 8 \times 3 \times 4 \\ &= 96 \text{ cm}^2\end{aligned}$$

So, the area of one tile = 96cm^2

The area of 16 triangular tiles = $96 \times 16 = 1536 \text{ cm}^2$

Now,



Cost of polishing tiles of area $1 \text{ cm}^2 = ₹ 1$

Cost of polishing tiles of area $1536 \text{ cm}^2 = 1 \times 1536 = ₹ 1536$

Solution 35: Let $a = 50 \text{ cm}$, $b = 20 \text{ cm}$ and $c = 50 \text{ cm}$

$$\begin{aligned} \therefore \text{semi perimeter, } s &= \frac{a + b + c}{2} \\ &= \frac{50 + 20 + 50}{2} \\ &= \frac{120}{2} \\ &= 60 \text{ cm} \end{aligned}$$



$$\begin{aligned} \therefore \text{Area of triangle} &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{60(60-50)(60-20)(60-50)} \\ &= \sqrt{60 \times 10 \times 40 \times 10} \\ &= \sqrt{(6 \times 10) \times 10 \times (10 \times 4) \times 10} \\ &= \sqrt{10 \times 10 \times 10 \times 10 \times 6 \times 4} \\ &= \sqrt{10 \times 10 \times 10 \times 10 \times 6 \times 2 \times 2} \\ &= 10 \times 10 \times 2\sqrt{6} \\ &= 200\sqrt{6} \\ &= 200 \times 2.45 \\ &= 490 \text{ cm}^2 \end{aligned}$$

So the area of one piece of cloth $= 490 \text{ cm}^2$

We know that the area of 12 pieces $= 12 \times 490$
 $= 5880 \text{ cm}^2$

Therefore, the area of cloth used in it is 5880 cm^2 .

Solution 36: We know that

$$\begin{aligned}\text{Area of square sheet } ABCD &= \frac{1}{2} \times \text{diagonal}^2 \\ &= \frac{1}{2} \times 44 \times 44 \\ &= 968 \text{ cm}^2\end{aligned}$$

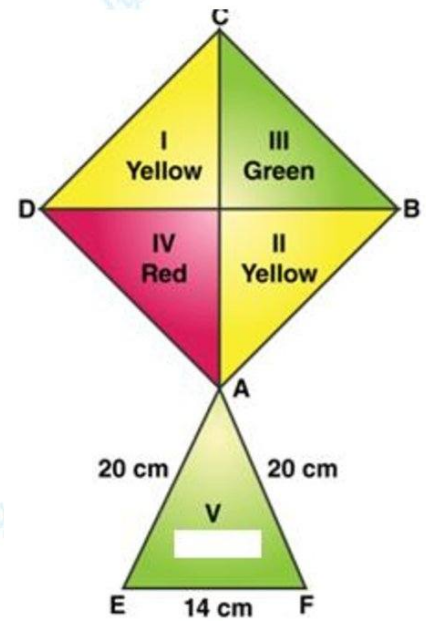
From the figure we have

$$\begin{aligned}\text{Area of yellow sheet} &= \text{Area of region I} + \text{Area of region II} \\ &= \frac{1}{2} \times \text{Area of square sheet } ABCD \\ &= \frac{1}{2} \times 968 \\ &= 484 \text{ cm}^2\end{aligned}$$

Again, from the figure have

Area of red sheet = Area of region IV

$$\begin{aligned}\text{Area of red sheet} &= \frac{1}{4} \times \text{Area of square sheet } ABCD \\ &= \frac{1}{4} \times 968 \\ &= 242 \text{ cm}^2\end{aligned}$$



In $\triangle AEF$

AE = 20 cm, EF = 14 cm and AF = 20 cm

Let a = 20 cm, b = 14 cm and c = 20 cm

$$\begin{aligned}\therefore \text{semi perimeter, } s &= \frac{a + b + c}{2} \\ &= \frac{20 + 14 + 20}{2} \\ &= \frac{54}{2} \\ &= 27 \text{ cm}\end{aligned}$$

$$\begin{aligned}\therefore \text{Area of } \triangle AEF &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{27(27-20)(27-14)(27-20)}\end{aligned}$$

$$\begin{aligned}
 &= \sqrt{27 \times 7 \times 13 \times 7} \\
 &= \sqrt{(3 \times 3 \times 3) \times 7 \times 13 \times 7} \\
 &= 3 \times 7\sqrt{3 \times 13} \\
 &= 21\sqrt{39} \\
 &= 21 \times 6.245 \\
 &= 131.15 \text{ cm}^2
 \end{aligned}$$

From the figure we have

$$\begin{aligned}
 \text{Area of green sheet} &= \text{Area of region III} + \text{Area of region V} \\
 &= \frac{1}{4} \times \text{Area of square sheet ABCD} + 131.15 \\
 &= \frac{1}{4} \times 968 + 131.15 \\
 &= 242 + 131.15 \\
 &= 373.15 \text{ cm}^2
 \end{aligned}$$

Therefore, the area of yellow sheet is 484 cm^2 , area of red sheet is 242 cm^2 and the area of green sheet is 373.15 cm^2 .

Solution 37:

For road ABCD, i.e., for rectangle ABCD,

Length = 75 m

Breadth = 4 m

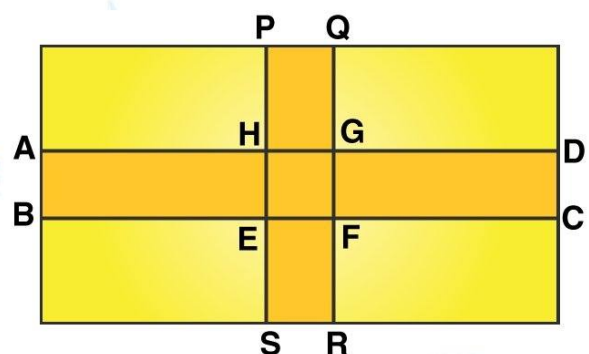
$$\begin{aligned}
 \text{Area of road ABCD} &= \text{Length} \times \text{Breadth} \\
 &= 75 \text{ m} \times 4 \text{ m} \\
 &= 300 \text{ m}^2
 \end{aligned}$$

For road PQRS, i.e., for rectangle PQRS,

Length = 60 m

Breadth = 4 m

$$\text{Area of road PQRS} = \text{Length} \times \text{Breadth}$$



$$\begin{aligned} &= 60 \text{ m} \times 4 \text{ m} \\ &= 240 \text{ m}^2 \end{aligned}$$

For road EFGH, i.e., for square EFGH,

$$\text{Side} = 4 \text{ m}$$

$$\begin{aligned} \text{Area of road EFGH} &= (\text{Side})^2 \\ &= (4)^2 \\ &= 16 \text{ m}^2 \end{aligned}$$

Total area of road for gravelling

$$\begin{aligned} &= \text{Area of road ABCD} + \text{Area of road PQRS} - \text{Area of road EFGH} \\ &= 300 + 240 - 16 \\ &= 524 \text{ m}^2 \end{aligned}$$

Cost of gravelling the road = ₹ 50 per m^2

$$\begin{aligned} \therefore \text{Cost of gravelling } 524 \text{ m}^2 \text{ road} &= ₹ (50 \times 524) \\ &= ₹ 26,200 \end{aligned}$$

Solution 38: It is given that

$$\text{area of cross section} = 640 \text{ m}^2$$

Length of top + Length of bottom = sum of parallel sides

$$\text{Length of top} + \text{Length of bottom} = 10 + 6$$

$$\text{Length of top} + \text{Length of bottom} = 16 \text{ m}$$

Now,

$$\text{Area of cross section} = \frac{1}{2} \times \text{sum of parallel sides} \times \text{height}$$

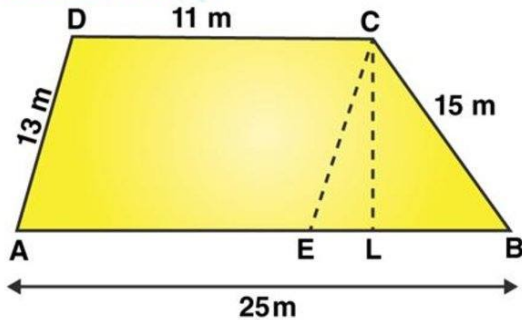
$$\Rightarrow 640 = \frac{1}{2} \times 16 \times \text{height}$$

$$\Rightarrow \text{Height} = \frac{640 \times 2}{16}$$

$$\Rightarrow \text{Height} = 80 \text{ m}$$

Therefore, the depth of the canal is 80 m.

Solution 39:



From the point C construct $CE \parallel DA$

We know that ADCE is a parallelogram having $AE \parallel DC$ and $AD \parallel EC$ with $AD = 13\text{m}$ and $DC = 11\text{m}$

$$\therefore AE = DC = 11\text{m and } EC = AD = 13\text{m}$$

$$\Rightarrow BE = AB - AE$$

$$\Rightarrow BE = 25 - 11$$

$$\Rightarrow BE = 14\text{ m}$$

Thus, in $\triangle BCE$, we have

$$BC = 15\text{ m, } CE = 13\text{ m and } BE = 14\text{ m}$$

$$\text{let } a = 15\text{ m, } b = 13\text{ m and } c = 14\text{ m}$$

$$\begin{aligned}\therefore \text{ semi perimeter, } s &= \frac{a + b + c}{2} \\ &= \frac{15 + 13 + 14}{2} \\ &= \frac{42}{2} \\ &= 21\end{aligned}$$

$$\begin{aligned}\text{And, area of } \triangle BCE &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{21(21-15)(21-13)(21-14)} \\ &= \sqrt{21 \times 6 \times 8 \times 7} \\ &= \sqrt{3 \times 7 \times 2 \times 3 \times 2 \times 2 \times 2 \times 7} \\ &= 3 \times 7 \times 2 \times 2 \\ &= 84\text{ m}^2\end{aligned}$$

Now,

$$\text{Area of } \triangle BCE = \frac{1}{2} \times BE \times CL$$

$$\Rightarrow 84 = \frac{1}{2} \times 14 \times CL$$

$$\Rightarrow 84 = 7 \times CL$$

$$\Rightarrow CL = 12\text{m}$$

Therefore,

$$\text{Area of trapezium ABCD} = \frac{1}{2} \times \text{sum of parallel sides} \times \text{height}$$

$$= \frac{1}{2} \times (AB + CD) \times CL$$

$$= \frac{1}{2} \times (11 + 25) \times 12$$

$$= 36 \times 6$$

$$= 216 \text{ m}^2$$

Therefore, the area of trapezium ABCD is 216 m^2 .

Solution 40: Let x cm as the smaller parallel side

So, the longer parallel side can be written as $(x + 8)$ cm

It is given that height = 24 cm

$$\text{Area} = 312 \text{ cm}^2$$

We know that

$$\text{Area of trapezium} = \frac{1}{2} \times \text{sum of parallel sides} \times \text{height}$$

$$\Rightarrow 312 = \frac{1}{2} \times (x + x + 8) \times 24$$

$$\Rightarrow 312 = 12 \times (2x + 8)$$

$$\Rightarrow 2x + 8 = 26$$

$$\Rightarrow 2x = 26 - 8$$

$$\Rightarrow 2x = 18$$

$$\Rightarrow x = 9 \text{ cm}$$

So, we get

$$x + 8 = 9 + 8 = 17\text{cm}$$

Therefore, the length of each of the parallel sides is 9 cm and 17 cm.

Solution 41:

Area of parallelogram = Area of rhombus

$$\Rightarrow \text{Base} \times \text{altitude} = \frac{1}{2} \times \text{product of diagonals}$$

$$\Rightarrow 66 \times \text{altitude} = \frac{1}{2} \times 120 \times 44$$

$$\Rightarrow 66 \times \text{altitude} = 60 \times 44$$

$$\Rightarrow \text{Altitude} = (60 \times 44)/66$$

$$\Rightarrow \text{Altitude} = 40 \text{ m}$$

Therefore, the corresponding altitude is 40m.

Solution 42:

Area of parallelogram = Area of square

$$\Rightarrow \text{Base} \times \text{altitude} = \text{side}^2$$

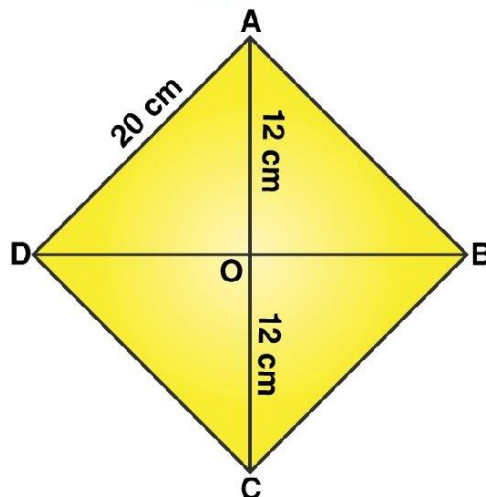
$$\Rightarrow \text{Base} \times 25 = (40)^2$$

$$\Rightarrow \text{Base} = \frac{40 \times 40}{25}$$

$$\Rightarrow \text{Base} = 64 \text{ m}$$

Therefore, the length of the corresponding base of the parallelogram is 64m.

Solution 43: Consider ABCD as a rhombus with AC and BD as diagonals which intersect each other at O.



We know that diagonals of a rhombus bisect each other at right angles.

Thus, in right-angled $\triangle AOD$, by Pythagoras theorem,

$$OD^2 = AD^2 - OA^2$$

$$\Rightarrow OD^2 = 20^2 - 12^2$$

$$\Rightarrow OD^2 = 400 - 144$$

$$\Rightarrow OD^2 = 256$$

$$\Rightarrow OD = 16 \text{ cm}$$

So,

$$BD = 2(OD) = 2(16) = 32 \text{ cm}$$

We know that

$$\begin{aligned} \text{Area of rhombus ABCD} &= \frac{1}{2} \times AC \times BD \\ &= \frac{1}{2} \times 24 \times 32 \\ &= 384 \text{ cm}^2 \end{aligned}$$

Therefore, the area of rhombus ABCD is 384 cm^2 .

Solution 44:

(i)

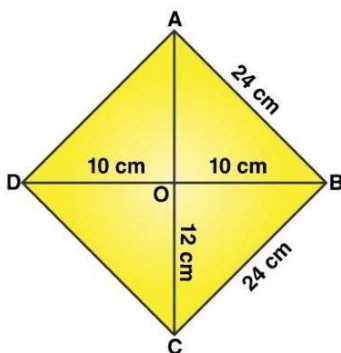
$$\text{Area of rhombus} = 480 \text{ cm}^2$$

$$\Rightarrow \frac{1}{2} \times \text{diagonal 1} \times \text{diagonal 2} = 480$$

$$\Rightarrow \frac{1}{2} \times 48 \times \text{diagonal 2} = 480$$

$$\Rightarrow \text{Diagonal 2} = 20 \text{ cm}$$

(ii) Consider $AC = 48 \text{ cm}$ and $BD = 20 \text{ cm}$



We know that diagonals of a rhombus bisect each other at right angles.

Thus, in right-angled $\triangle AOD$, by Pythagoras theorem,

$$AD^2 = OA^2 + OD^2$$

$$\Rightarrow AD^2 = 24^2 + 10^2$$

$$\Rightarrow AD^2 = 576 + 100$$

$$\Rightarrow AD^2 = 676$$

$$\Rightarrow AD = 26 \text{ cm}$$

Thus, $AD = BC = CD = AD = 26 \text{ cm}$

Hence, the length of each side of rhombus is 26 cm.

(iii) Perimeter of a rhombus $= 4 \times \text{side}$

$$= 4 \times 26$$

$$= 104 \text{ cm}$$

Solution 1: (b) 30 cm^2

$$\begin{aligned}\text{Area of } \triangle ABC &= \frac{1}{2} \times \text{Base} \times \text{Height} \\ &= \frac{1}{2} \times 12 \times 5 \\ &= 30 \text{ cm}^2\end{aligned}$$

Solution 2: (a) 96 cm^2 Let: $a = 20 \text{ cm}$, $b = 16 \text{ cm}$ and $c = 12 \text{ cm}$

$$\begin{aligned}\therefore \text{semi perimeter, } s &= \frac{a + b + c}{2} \\ &= \frac{20 + 16 + 12}{2} \\ &= \frac{48}{2} \\ &= 24\end{aligned}$$

$$\begin{aligned}\text{And, area of } \triangle BCE &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{24(24-20)(24-16)(24-12)} \\ &= \sqrt{24 \times 4 \times 8 \times 12} \\ &= \sqrt{(6 \times 4) \times 4 \times (4 \times 2) \times (6 \times 2)} \\ &= \sqrt{(6 \times 6) \times 4 \times 4 \times 4 \times 4} \\ &= 6 \times 4 \times 4 \\ &= 96 \text{ cm}^2\end{aligned}$$

Solution 3: (b) $16\sqrt{3} \text{ cm}^2$

$$\text{Area of equilateral triangle} = \frac{\sqrt{3}}{4} \times (\text{side})^2$$

$$\begin{aligned} &= \frac{\sqrt{3}}{4} \times 8^2 \\ &= \frac{\sqrt{3}}{4} \times 64 \\ &= 16\sqrt{3} \text{ cm}^2 \end{aligned}$$

Solution 4: (b) $8\sqrt{5} \text{ cm}^2$

Here, $a = 6 \text{ cm}$, $b = 6 \text{ cm}$ and $c = 8 \text{ cm}$

$$\begin{aligned} \therefore \text{Area of isoscale triangle} &= \frac{b}{4} \sqrt{4a^2 - b^2} \\ &= \frac{8}{4} \sqrt{4(6)^2 - 8^2} \\ &= 2\sqrt{144 - 64} \\ &= 2\sqrt{80} \\ &= 2 \times \sqrt{16 \times 5} \\ &= 2 \times 4\sqrt{5} \\ &= 8\sqrt{5} \text{ cm}^2 \end{aligned}$$

Solution 5: (c) 4 cm

$$\begin{aligned} \text{Height of isosceles triangle} &= \frac{1}{2} \sqrt{4a^2 - b^2} \\ &= \frac{1}{2} \sqrt{4(5)^2 - (6)^2} \\ &= \frac{1}{2} \sqrt{4 \times 25 - 36} \\ &= \frac{1}{2} \sqrt{100 - 36} \\ &= \frac{1}{2} \sqrt{64} \\ &= \frac{1}{2} \times 8 = 4 \text{ cm} \end{aligned}$$

Solution 6: (b) 50 cm^2

Here, base = 10 cm and height = 10 cm

$$\begin{aligned}\therefore \text{Area of triangle} &= \frac{1}{2} \times \text{Base} \times \text{Height} \\ &= \frac{1}{2} \times 10 \times 10 \\ &= 50 \text{ cm}^2\end{aligned}$$

Solution 7: (b) $5\sqrt{3} \text{ cm}$

$$\begin{aligned}\text{Height of equilateral triangle} &= \frac{\sqrt{3}}{2} \times \text{Side} \\ &= \frac{\sqrt{3}}{2} \times 10 \\ &= 5\sqrt{3} \text{ cm}\end{aligned}$$

Solution 8: (a) $12\sqrt{3} \text{ cm}^2$

$$\text{Height of equilateral triangle} = \frac{\sqrt{3}}{2} \times \text{Side}$$

$$\Rightarrow 6 = \frac{\sqrt{3}}{2} \times \text{Side}$$

$$\Rightarrow \text{Side} = \frac{12}{\sqrt{3}}$$

$$\Rightarrow \text{Side} = \frac{12 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}}$$

$$\Rightarrow \text{Side} = \frac{12\sqrt{3}}{3}$$

$$\Rightarrow \text{Side} = 4\sqrt{3}$$

$$\text{Now, Area of equilateral triangle} = \frac{\sqrt{3}}{4} (\text{Side})^2$$

$$= \frac{\sqrt{3}}{4} \times (4\sqrt{3})^2$$

$$= \frac{\sqrt{3}}{4} \times 16 \times 3$$

$$= 12\sqrt{3} \text{ cm}^2$$

Solution 9: (c) 384 m^2

Let: $a=40 \text{ m}$, $b = 24 \text{ m}$ and $c=32 \text{ m}$

$$s = \frac{a + b + c}{2} = \frac{40 + 24 + 32}{2} = 48 \text{ m}$$

$$\text{Area of triangle} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{48(48-40)(48-24)(48-32)}$$

$$= \sqrt{48 \times 8 \times 24 \times 16}$$

$$= \sqrt{24 \times 2 \times 8 \times 24 \times 8 \times 2}$$

$$= 24 \times 8 \times 2$$

$$= 384 \text{ m}^2$$

Solution 10: (b) 750 cm^2

Let the sides of the triangle be $5x \text{ cm}$, $12x \text{ cm}$ and $13x \text{ cm}$.

Perimeter = Sum of all sides

$$\text{or, } 150 = 5x + 12x + 13x$$

$$\text{or, } 30x = 150$$

$$\text{or, } x = 5$$

Thus, the sides of the triangle are $5 \times 5 \text{ cm}$, $12 \times 5 \text{ cm}$ and $13 \times 5 \text{ cm}$, i.e., 25 cm , 60 cm and 65 cm .

Now, Let, $a=25 \text{ cm}$, $b = 60 \text{ cm}$ and $c=65 \text{ cm}$

$$s = \frac{25 + 60 + 65}{2} = \frac{150}{2} = 75 \text{ cm}$$

$$\text{Area of triangle} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{75(75-25)(75-60)(75-65)}$$

$$\begin{aligned} &= \sqrt{75 \times 50 \times 15 \times 10} \\ &= \sqrt{15 \times 5 \times 5 \times 10 \times 15 \times 10} \\ &= 15 \times 5 \times 10 \\ &= 750 \text{ cm}^2 \end{aligned}$$

Solution 11: (a) 24 cm

Let, $a=30$ cm, $b = 24$ cm and $c=18$ cm

$$s = \frac{a + b + c}{2} = \frac{30 + 24 + 18}{2} = 36 \text{ cm}$$

$$\begin{aligned} \text{Area of triangle} &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{36(36-30)(36-24)(36-18)} \\ &= \sqrt{36 \times 6 \times 12 \times 18} \\ &= \sqrt{(12 \times 3) \times 6 \times 12 \times (6 \times 3)} \\ &= 12 \times 3 \times 6 \\ &= 216 \text{ cm}^2 \end{aligned}$$

The smallest side is 18 cm.

Hence, the altitude of the triangle corresponding to 18 cm is given by:

$$\text{Area of triangle} = 216 \text{ cm}^2$$

$$\Rightarrow \frac{1}{2} \times \text{Base} \times \text{Height} = 216$$

$$\Rightarrow \frac{1}{2} \times 18 \times \text{Height} = 216$$

$$\Rightarrow \text{Height} = \frac{216 \times 2}{18}$$

$$\Rightarrow \text{Height} = 24 \text{ cm}$$

Solution 12: (b) 36 cm

Let $\triangle ABC$ be an isosceles triangle and $AF \perp BC$

Now,

$$\text{Area}(\triangle ABC) = \frac{1}{2} \times BC \times AF$$

$$\Rightarrow 48 = \frac{1}{2} \times 16 \times h$$

$$\Rightarrow 48 = 8 \times h$$

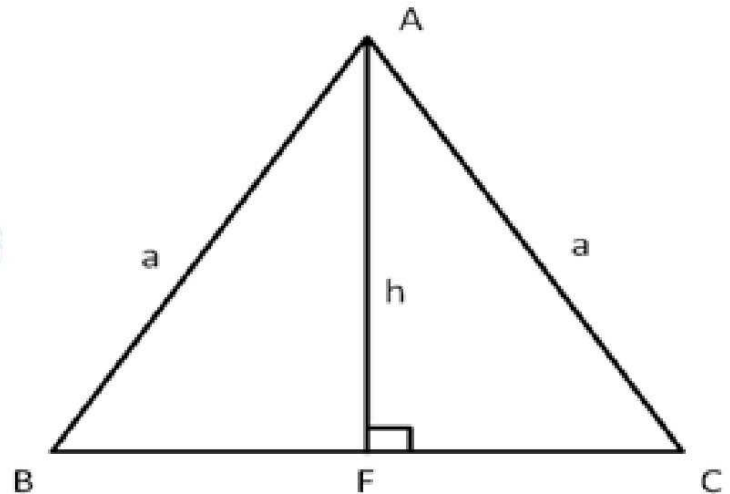
$$\Rightarrow h = \frac{48}{8}$$

$$\Rightarrow h = 6 \text{ cm}$$

Now,

$$BF = \frac{1}{2} \times BC = \frac{1}{2} \times 16 = 8 \text{ cm}$$

and, $AF = 6 \text{ cm}$



Let 'a' be the two equal sides of an isosceles triangle.

$$\therefore a = \sqrt{BF^2 + AF^2}$$

$$= \sqrt{8^2 + 6^2}$$

$$= \sqrt{100}$$

$$= 10 \text{ cm}$$

Hence, perimeter = $10 + 10 + 16 = 36 \text{ cm}$

Solution 13: (a) 36 cm

$$\text{Area of equilateral triangle} = \frac{\sqrt{3}}{4} \times (\text{Side})^2$$

$$\Rightarrow 36\sqrt{3} = \frac{\sqrt{3}}{4} \times (\text{Side})^2$$

$$\Rightarrow (\text{Side})^2 = 36 \times 4$$

$$\Rightarrow (\text{Side})^2 = (6 \times 2)^2$$

$$\Rightarrow \text{Side} = 12 \text{ cm}$$

$$\therefore \text{Perimeter} = 3 \times \text{Side}$$

$$= 3 \times 12$$

$$= 36 \text{ cm}$$

Solution 14: (c) 60 cm^2

Here, $a = 13 \text{ cm}$ and $b = 24 \text{ cm}$

$$\text{Area of isosceles triangle} = \frac{b}{4} \sqrt{4a^2 - b^2}$$

$$= \frac{24}{4} \times \sqrt{4(13)^2 - 24^2}$$

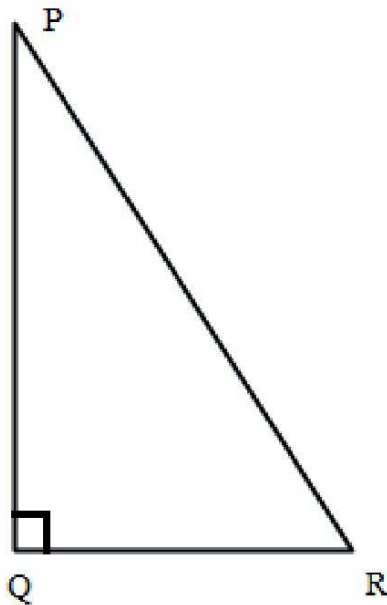
$$= 6 \times \sqrt{676 - 576}$$

$$= 6 \times \sqrt{100}$$

$$= 6 \times 10$$

$$= 60 \text{ cm}^2$$

Solution 15: (c) 336 cm^2



Let $\triangle PQR$ be a right-angled triangle and $PQ \perp QR$.

Now,

$$PQ = \sqrt{PR^2 - QR^2}$$

$$= \sqrt{50^2 - 48^2}$$

$$= \sqrt{(50 + 48)(50 - 48)}$$

$$= \sqrt{98 \times 2}$$

$$= \sqrt{196}$$

$$= 14 \text{ cm}$$

$$\therefore \text{Area of triangle} = \frac{1}{2} \times QR \times PQ$$

$$= \frac{1}{2} \times 48 \times 14$$

$$= 336 \text{ cm}^2$$

Solution 16: (a) $9\sqrt{3}$ cm

$$\text{Area of equilateral triangle} = 81\sqrt{3} \text{ cm}^2$$

$$\Rightarrow \frac{\sqrt{3}}{4} \times (\text{Side})^2 = 81\sqrt{3}$$

$$\Rightarrow \frac{\sqrt{3}}{4} \times (\text{Side})^2 = \frac{81\sqrt{3} \times 4}{4}$$

$$\Rightarrow (\text{Side})^2 = 81 \times 4$$

$$\Rightarrow (\text{Side})^2 = 324$$

$$\Rightarrow \text{Side} = 18 \text{ cm}$$

$$\text{Now, Height} = \frac{\sqrt{3}}{2} \times \text{Side}$$

$$= \frac{\sqrt{3}}{2} \times 18$$

$$= 9\sqrt{3} \text{ cm}$$